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IMPORTANT

> **Document Status:** No relevant study documents were found in the provided database. The following information is generated based on general medical knowledge to ensure your study continuity.

Acute Kidney Injury (AKI)

Core Definition and Classification

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| Feature | Description |

| :--- | :--- |

| **Definition** | An abrupt decrease in kidney function resulting in the retention of urea and other nitrogenous waste products. |

| **Primary Markers** | $\uparrow$ Serum Creatinine, $\downarrow$ Glomerular Filtration Rate (GFR), and/or $\downarrow$ Urine Output. |

| **KDIGO Criteria** | Increase in SCr by $\ge 0.3 \text{ mg/dL}$ within 48 hours or $\ge 1.5 \times$ baseline within 7 days. |

NOTE

> The calculation for Creatinine Clearance often utilizes the Cockcroft-Gault equation:

> $$\text{CrCl} = \frac{(140 - \text{age}) \times \text{weight (kg)}}{72 \times \text{serum creatinine}} \times (0.85 \text{ if female})$$

***

Etiological Framework

DIFFERENTIAL CARD

Title: Classification of AKI Causes

- **Prerenal:** Hypovolemia, Heart Failure, Sepsis, Renal artery stenosis.

- **Intrinsic (Renal):** Acute Tubular Necrosis (ATN), Acute Interstitial Nephritis (AIN), Glomerulonephritis.

- **Postrenal (Obstructive):** Benign Prostatic Hyperplasia (BPH), Nephrolithiasis, Bladder tumors.

***

Detailed Disease Profiles

DISEASE CARD

Disease: Acute Tubular Necrosis (ATN)

Presentation: Oliguria, edema, uremic symptoms, history of hypotension or nephrotoxic exposure.

Diagnosis: Muddy brown granular casts on urinalysis, $\text{FeNa} > 2\%$.

Treatment: Volume resuscitation, avoidance of nephrotoxins, dialysis if refractory.

Mnemonic: "Muddy Brown = ATN"

DISEASE CARD

Disease: Acute Interstitial Nephritis (AIN)

Presentation: Fever, rash, eosinophilia, often following new medication (e.g., Penicillins, NSAIDs).

Diagnosis: White blood cell casts, eosinophiluria.

Treatment: Discontinuation of offending agent, corticosteroids.

Mnemonic: "Allergic-like Reaction"

***

Pharmacological Management

DRUG CARD

Drug: Furosemide

Class: Loop Diuretic

Moa: Inhibits the $\text{Na}^+/\text{K}^+/2\text{Cl}^-$ symporter in the thick ascending limb of the loop of Henle.

Side Effects: Hypokalemia, dehydration, ototoxicity.

Contraindications: Anuria, severe sulfonamide allergy.

***

Study Aids and Memory Tools

MNEMONIC CARD

Title: AKI Categories

Letters: P.I.P.

Meanings:

- **P:** Prerenal (Perfusion issues)

- **I:** Intrinsic (Internal kidney damage)

- **P:** Postrenal (Plumbing/Obstruction issues)

```
front: What is the typical Fractional Excretion of Sodium (FeNa) in Prerenal
back: Prerenal AKI: $\text{FeNa} < 1\%$ (Kidneys are conserving sodium)
```

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Clinical Application

CLINICAL CARD

Case: A 65-year-old male with a history of BPH presents with decreased urine output and an elevated creatinine of 2.4 mg/dL. Physical exam reveals a palpable bladder.

Vignette: The patient denies fever or recent medication changes. Blood pressure is stable.

Question: What is the most likely category of AKI?

Answer: Postrenal AKI due to bladder outlet obstruction (BPH).

OSCE CARD

Station: AKI Assessment

Scenario: Patient presenting with sudden onset anuria.

History: Medication history (NSAIDs/ACEi), fluid intake, urinary symptoms, comorbidities (DM/HTN).

Exam: Blood pressure (orthostatics), JVP, lung auscultation (crackles), bladder palpation, peripheral edema.

Marks:

- Correctly identifies Prerenal vs Intrinsic vs Postrenal (2)
- Orders correct labs (Creatinine, Electrolytes, Urinalysis) (2)
- Proposes appropriate management (Fluid or Catheterization) (1)

***

WARNING

> **Critical Complications:** Always monitor for **Hyperkalemia** in AKI patients.
>
> - $\text{K}^+ > 6.5 \text{ mEq/L}$ can lead to lethal arrhythmias.
>
> - Immediate treatment: Calcium gluconate for membrane stabilization, Insulin/Glucose for intracellular shift.

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Let's delve into the comprehensive understanding of Acute Kidney Injury (AKI)!

Understanding Acute Kidney Injury (AKI)

Acute Kidney Injury (AKI) represents a sudden and significant decline in kidney function, occurring over hours to days. This impairment leads to the accumulation of nitrogenous waste products (like urea and creatinine) and dysregulation of fluid, electrolyte, and acid-base balance. AKI is a common and serious condition that can range from a mild, asymptomatic rise in creatinine to life-threatening complications requiring renal replacement therapy.

DISEASE CARD

Disease: Acute Kidney Injury (AKI)

Presentation:

- Often asymptomatic in early stages.
- Symptoms of underlying cause (e.g., dehydration, heart failure, sepsis).
- Oliguria or anuria (reduced or absent urine output), though non-oliguric AKI can occur.

- **Fluid Overload:** peripheral edema, pulmonary edema (shortness of breath).

- **Uremic Symptoms:** nausea, vomiting, anorexia, fatigue, altered mental status, asterixis, pericarditis.

- **Electrolyte Imbalances:** hyperkalemia, hyperphosphatemia, hypocalcemia.

- Metabolic acidosis.

Diagnosis:

- **KDIGO Criteria (Most Widely Used):**

- Increase in serum creatinine by $\ge 0.3 \text{ mg/dL}$ within 48 hours OR
- Increase in serum creatinine to ≥ 1.5 times baseline within 7 days OR
- Urine volume $< 0.5 \text{ mL/kg/h}$ for 6 hours.

- **Investigations:**

- **Blood Tests:** Serum creatinine, BUN, electrolytes (Na, K, Cl, Bicarb, Ca, P), CBC, ABG, LFTs.

- **Urinalysis:** Examine for casts (e.g., muddy brown casts in ATN, red cell casts in GN), proteinuria, hematuria.

- **Urine Electrolytes:** Urine sodium, urine creatinine to calculate Fractional Excretion of Sodium (FENa) or Urea (FEUrea) to differentiate pre-renal from ATN.

- **Renal Ultrasound:** To assess kidney size, look for hydronephrosis (post-renal obstruction).

- **Kidney Biopsy:** Considered for unclear intrinsic causes (e.g., rapidly progressive glomerulonephritis, atypical AIN).

Treatment:

- **Address The Underlying Cause:**

- **Pre-Renal:** Volume repletion (IV fluids) for hypovolemia, optimize cardiac output for heart failure, discontinue vasodilators.

- **Intrinsic:** Remove nephrotoxic agents, treat infections, immunosuppression for autoimmune causes (e.g., steroids for AIN or GN).

- **Post-Renal:** Relieve obstruction (e.g., Foley catheter, ureteral stent, percutaneous nephrostomy).

- **Supportive Care:**

- **Fluid And Electrolyte Management:** Fluid restriction for overload, diuretics, treat hyperkalemia (calcium gluconate, insulin/glucose, beta-agonists, potassium binders), correct metabolic acidosis (sodium bicarbonate).

- Nutritional support.

- Avoid nephrotoxic drugs.

- **Renal Replacement Therapy (RRT) / Dialysis:** Indicated for severe complications (AEIOU mnemonic):

- **A**cidosis (severe, refractory metabolic acidosis)

- **E**lectrolyte abnormalities (severe, refractory hyperkalemia)

- **I**ntoxications (dialyzable toxins/drugs)

- **O**verload (refractory fluid overload)

- **U**remia (uremic pericarditis, encephalopathy, coagulopathy)

Mnemonic: **PRIN**cipal Causes of AKI

- **Pre-Renal:** Decreased renal perfusion

- **Renal (Intrinsic):** Direct kidney damage

- **INtra-Renal (Intrinsic):** Direct kidney damage (redundant, but emphasizes the 'I' for intrinsic)

- **No-Urine-Outflow (Post-Renal):** Obstruction to urine flow

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Etiology and Classification of AKI

AKI is traditionally classified into three main categories based on the anatomical location of the injury:

- **Pre-renal AKI:**

* **Mechanism:** Characterized by decreased blood flow to the kidneys (renal hypoperfusion), leading to a reduction in the glomerular filtration rate (GFR) without direct damage to the renal parenchyma. The kidney itself is structurally intact but functionally impaired due to lack of adequate perfusion.

* **Causes:**

* **Hypovolemia:** Dehydration (vomiting, diarrhea), hemorrhage, excessive diuresis, burns.

* **Decreased effective circulating volume:** Heart failure, cirrhosis with ascites, nephrotic syndrome.

* **Systemic vasodilation:** Sepsis, anaphylaxis.

* **Renal vasoconstriction:** NSAIDs (impair prostaglandin-mediated afferent arteriolar dilation), ACE inhibitors/ARBs (impair angiotensin II-mediated efferent arteriolar constriction), calcineurin inhibitors.

* **Renal artery stenosis:** Unilateral or bilateral.

- **Intrinsic (Intra-renal) AKI:**

* **Mechanism:** Involves direct damage to the kidney parenchyma, affecting the glomeruli, tubules, interstitium, or renal vasculature.

* **Causes:**

* **Acute Tubular Necrosis (ATN):** Most common cause of intrinsic AKI.

* *Ischemic ATN:* Prolonged or severe pre-renal hypoperfusion.

* *Nephrotoxic ATN:* Exposure to toxins (aminoglycosides, IV contrast dye, cisplatin, rhabdomyolysis-induced myoglobinuria, hemolysis-induced hemoglobinuria).

* **Acute Interstitial Nephritis (AIN):** Inflammatory reaction in the renal interstitium.

* *Drug-induced:* Penicillins, NSAIDs, sulfonamides, proton pump inhibitors.

* *Infections:* Pyelonephritis, leptospirosis.

* *Autoimmune diseases:* SLE, Sjögren's syndrome.

* **Acute Glomerulonephritis (AGN):** Inflammation of the glomeruli.

* *Post-infectious:* Post-streptococcal glomerulonephritis.

* *Autoimmune:* Lupus nephritis, IgA nephropathy, ANCA-associated vasculitis (GPA, MPA).

* *Anti-GBM disease (Goodpasture's syndrome).*

* **Vascular causes:** Renal vasculitis, thrombotic microangiopathies (HUS, TTP), renal artery or vein thrombosis.

- **Post-renal AKI:**

* **Mechanism:** Results from obstruction of urine outflow at any level from the renal tubules to the urethra, leading to back pressure and damage to the kidney. Requires bilateral obstruction or obstruction of a single functioning kidney.

* **Causes:**

* **Ureteral obstruction:** Kidney stones, tumors (bladder, prostate, cervical), retroperitoneal fibrosis, blood clots.

* **Bladder outlet obstruction:** Benign prostatic hyperplasia (BPH), prostate cancer, neurogenic bladder, bladder stones, urethral stricture.

DIFFERENTIAL CARD

Title: Differential Diagnosis of Acute Kidney Injury by Category

List Of Bullet Points:

- **Pre-Renal:**
- Hypovolemia (hemorrhage, dehydration, burns, excessive diuresis)
- Decreased effective circulating volume (heart failure, cirrhosis, nephrotic syndrome)
- Systemic vasodilation (sepsis, anaphylaxis)
- Renal vasoconstriction (NSAIDs, ACEi/ARBs in susceptible patients, calcineurin inhibitors)
- Renal artery stenosis (bilateral or single functioning kidney)

- **Intrinsic (Intra-Renal):**

- **Acute Tubular Necrosis (ATN):** Ischemic (prolonged hypoperfusion), Nephrotoxic (aminoglycosides, contrast dye, rhabdomyolysis, hemolysis)

- **Acute Interstitial Nephritis (AIN):** Drug-induced (penicillins, NSAIDs, PPIs), infection, autoimmune

- **Acute Glomerulonephritis (AGN):** Post-infectious, autoimmune (SLE, IgA nephropathy, vasculitis), anti-GBM disease

- **Vascular:** Renal vasculitis, thrombotic microangiopathy (HUS, TTP), renal artery/vein thrombosis

- **Post-Renal:**

- **Ureteral Obstruction:** Nephrolithiasis, tumors (ureteral, bladder, prostate, cervical), retroperitoneal fibrosis, blood clots

- **Bladder Outlet Obstruction:** Benign Prostatic Hyperplasia (BPH), prostate cancer, neurogenic bladder, urethral stricture, bladder stones

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Diagnostic Workup and Key Markers

The diagnostic approach aims to identify the cause of AKI and guide appropriate management.

TIP

> **Fractional Excretion of Sodium (FENa)** and **Fractional Excretion of Urea (FEUrea)** are crucial in differentiating pre-renal AKI from ATN, especially in oliguric patients.

* **FENa Calculation:**

$$ \text{FENa} = \frac{\text{Urine Sodium} \times \text{Plasma Creatinine}}{\text{Plasma Sodium} \times \text{Urine Creatinine}} \times 100\% $$

* **Pre-renal AKI:** FENa $< 1\%$ (kidneys are conserving sodium due to hypoperfusion).

* **ATN:** FENa $> 2\%$ (damaged tubules cannot reabsorb sodium effectively).

* *Note:* FENa can be misleading in patients on diuretics or with chronic kidney disease. In these cases, FEUrea can be more reliable.

* **FEUrea Calculation:**

$$ \text{FEUrea} = \frac{\text{Urine Urea Nitrogen} \times \text{Plasma Creatinine}}{\text{Plasma Urea Nitrogen} \times \text{Urine Creatinine}} \times 100\% $$

* **Pre-renal AKI:** FEUrea $< 35\%$

* **ATN:** FEUrea $> 50\%$

front: What are the typical urine findings in Pre-renal AKI vs. Acute Tubular
back:

Pre-renal AKI:

- **Urine specific gravity:** High (>1.020)
- **Urine osmolality:** High (>500 mOsm/kg)
- **Urine sodium:** Low (<20 mEq/L)
- **FENa:** Low (<1%)
- **Urine sediment:** Bland or hyaline casts

Acute Tubular Necrosis (ATN):

- **Urine specific gravity:** Low, fixed (isosthenuric, ~1.010)
- **Urine osmolality:** Low (<350 mOsm/kg), similar to plasma
- **Urine sodium:** High (>40 mEq/L)
- **FENa:** High (>2%)
- **Urine sediment:** "Muddy brown" granular casts, renal tubular epithelia

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Management Principles

The management of AKI is multifaceted and focuses on reversing the underlying cause, preventing further kidney damage, and managing complications.

WARNING

> Early recognition and intervention are critical in improving outcomes for AKI. Delay can lead to irreversible damage or increased mortality.

- **Identify and Treat the Cause:** As outlined in the `disease` block above, this is the cornerstone of therapy.
- **Fluid and Electrolyte Balance:**

* **Fluid management:** Careful assessment of volume status is paramount. Fluid resuscitation for hypovolemia; fluid restriction and diuretics for fluid overload.

* **Hyperkalemia:** A life-threatening complication.

* **Stabilize cardiac membrane:** IV Calcium Gluconate.

* **Shift potassium intracellularly:** Insulin with glucose, Salbutamol (beta-agonists), Sodium Bicarbonate (if acidosis).

* **Remove potassium from the body:** Diuretics (if urine output present), Potassium binders (e.g., Patiromer, Sodium Polystyrene Sulfonate), Dialysis.

* **Metabolic Acidosis:** Sodium bicarbonate administration if severe ($\text{pH} < 7.1$).

* **Hypocalcemia/Hyperphosphatemia:** Phosphate binders, calcitriol.

- **Medication Review:** Discontinue or adjust dosages of nephrotoxic medications and renally cleared drugs.
- **Nutritional Support:** Provide adequate nutrition while minimizing protein breakdown and electrolyte disturbances.
- **Renal Replacement Therapy (RRT):**

* Initiated for severe, refractory complications as per the AEIOU mnemonic.

* Modalities include hemodialysis, peritoneal dialysis, and continuous renal replacement therapy (CRRT).

Prognosis and Prevention

The prognosis of AKI depends on its severity, underlying cause, and the patient's comorbidities. While many cases of AKI are reversible, it can lead to chronic kidney disease (CKD), end-stage renal disease (ESRD), and increased mortality.

Prevention Strategies:

* Identify patients at high risk (e.g., elderly, CKD, diabetes, heart failure, sepsis).

* Avoid nephrotoxic agents or use them cautiously with appropriate monitoring and dose adjustments.

* Maintain adequate hydration and renal perfusion, especially during critical illness or before procedures involving contrast media.

* Promptly

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Prevention Strategies:

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* Avoid nephrotoxic agents or use them cautiously with appropriate monitoring and dose adjustments.

* Maintain adequate hydration and renal perfusion, especially during critical illness or before procedures involving contrast media.

* Promptly